Experiment- 3

Aim

To perform memory analysis using Volatility tool:

1. To list all running processes from a memory image
2. To list all network connections from a memory image
3. To analyze firewall status from a memory image

Tools Required

- Volatility (Volatility 2 / Volatility 3)
- Memory Image File (.raw / .mem)
- Python Environment
- Computer System

Theory

Memory forensics (RAM analysis) is a technique used to analyze the contents of volatile memory. It helps investigators detect running processes, network activity, and system configuration at the time of capture.

Volatility is an advanced open-source tool used for extracting digital artifacts from memory images. It supports multiple operating systems and helps in identifying malicious activities.

Using Volatility, we can:

- Identify running processes
- Analyze network connections
- Check firewall and security configurations

Prerequisites

```
# Install Volatility 3 (recommended for modern systems)
pip3 install volatility3

# Or use Volatility 2 (for older memory images)
git clone https://github.com/volatilityfoundation/volatility.git
cd volatility
python2 setup.py install
```

Procedure

Step 1: Profile Identification

Command:

```
vol -f memory_image.raw imageinfo
```

Result:

- OS profile (e.g., Win10x64) is identified

```
Volatility 3 Framework 2.x.x

PID     PPID    ImageFileName      Threads  Handles  SessionId  Wow64   CreateTime
--------------------------------------------------------------------------------
4       0       System             92       500      -          False   2024-01-15 10:00:00
320     4       smss.exe           2        30       -          False   2024-01-15 10:00:01
456     320     csrss.exe          10       200      0          False   2024-01-15 10:00:02
512     320     wininit.exe        3        80       0          False   2024-01-15 10:00:02
600     512     services.exe       8        150      0          False   2024-01-15 10:00:03
650     512     lsass.exe          7        120      0          False   2024-01-15 10:00:03
900     600     svchost.exe        12       300      0          False   2024-01-15 10:00:04
1200    900     chrome.exe         25       600      1          True    2024-01-15 10:05:00
```

Step 2: List Running Processes

Command (Volatility 3):

vol -f memory_image.raw windows.pslist

Command (Volatility 2):

volatility -f memory_image.raw --profile=Win10x64 pslist

Sample Output:

PID PPID ProcessName

---

4 0 System

320 4 smss.exe

600 512 services.exe

900 600 svchost.exe

1200 900 chrome.exe

Explanation

- Its show System running processes list
- Its identify Suspicious process
- Parent-child relation important

Step 3: List Network Connections

Command (Volatility 3):

vol -f memory_image.raw windows.netscan

Command (Volatility 2):

volatility -f memory_image.raw --profile=Win10x64 netscan

Sample Output:

Proto Local Address Foreign Address State PID Process

TCP 192.168.1.10 142.250.183.78 ESTABLISHED 1200 chrome.exe

TCP 0.0.0.0:135 0.0.0.0 LISTENING 772 RpcSs

UDP 0.0.0.0:68 *:* - 900 svchost.exe

Explanation

- Will display all Network connections
- Will identify External IP connections
- Will identify Malware activity detect

Step 4: Firewall Status Analysis

Method 1: Check Firewall Service

vol -f memory_image.raw windows.pslist | grep -i mpssvc

Method 2: Registry Analysis

vol -f memory_image.raw windows.registry.printkey --key "HKEY_LOCAL_MACHINE\\SOFTWARE\\Policies\\Microsoft\\WindowsFirewall\\DomainProfile"

Output:

EnableFirewall REG_DWORD 1

Interpretation

- 1 → Firewall Enabled
- 0 → Firewall Disabled

Result

- Running processes listed successfully
- Network connections identified successfully
- Firewall status analyzed successfully